Assessing Tick-Borne Pathogen Transmission Risk in Pennsylvania Using Machine Learning and Geospatial Analysis

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Abstract

Tick-borne diseases are a growing public health concern, due to an increasing vector habitat range in Pennsylvania. This study uses machine learning and ecological data modeling to understand how environmental factors, tick life stages, and geographic patterns influence the distribution of ticks and tick-borne pathogens. Results show that biological factors such as tick species and life stage are stronger predictors of pathogen presence. Spatial analysis reveals hotspots of pathogen prevalence, while seasonal trends further influence risk patterns in PA. Machine learning (ML) models were effective in identifying key disease risk predictors and their patterns. These findings highlight the importance of combining ecological and data driven approaches to better predict disease risk and support public health efforts to mitigate potential disease outbreaks.

Objectives

- ❑ Identify environmental and ecological factors associated with pathogen presence in Pennsylvania tick populations
- ❑ Analyze how tick species and life stage influence the likelihood of pathogen presence
- ❑ Determine geographic patterns and hotspots of tick pathogen prevalence across Pennsylvania
- ❑ Apply machine learning and geospatial analysis to model and predict pathogen distribution and disease risk

Research Questions

1. How do environmental and ecological factors influence tick pathogen distribution in Pennsylvania, and how can machine learning be used to predict these patterns ?
2. What spatial patterns and environmental conditions define hotspots of tick pathogen prevalence across Pennsylvania?

Existing Work

- ❑ Estrada et al. [1] identified that climate, vegetation, and land use influence tick-borne pathogen patterns.
- ❑ Machine learning can predict tick distribution and pathogen risk by combining environmental and ecological data [2].
- ❑ Tick species and life stage affect infection rates, with adult ticks more likely to carry pathogens [2].
- ❑ Combining spatial, environmental, and biological data improves overall prediction accuracy [3].
- ❑ Geographic clustering leads to localized hotspots of pathogen prevalence [4].

Methods

Tick surveillance data were collected from statewide Pennsylvania dataset [5] and **included tick species, life stage, geographic location, temperature, season or month, and pathogen presence or absence.** All preprocessing, statistical analysis, geospatial mapping, and machine learning were performed in Python. To examine how environmental and spatial factors relate to infection status, **correlation analysis** was first performed using parameters **temperature**, **distance_m**, **latitude**, **longitude**, and **month**. **Pathogen presence** had only weak individual correlations with these variables (Fig 1), with the strongest positive relationships observed for **month (0.21)** and **distance_m (0.19)**, suggesting that infection risk is influenced by multiple interacting factors rather than one dominant variable. Consequently, four machine learning models, Logistic Regression, Random Forest, Gradient Boosting, and XGBoost, were trained using the same predictor set, specifically **temperature**, **month**, **latitude**, **longitude**, **nymph**, **adult female**, and **adult male**, and evaluated using **10-fold stratified cross validation**. Model performance was compared using **Accuracy, Precision, Recall, Specificity, F1 Score, ROC AUC**, and **MCC**, as summarized in **Table 1**. Among the models tested, **Gradient Boosting** showed the strongest overall performance, achieving the best balance across the major evaluation metrics, and was therefore selected as the final model for predicting **pathogen presence.**

Table 1: Evaluation Metrics of all models

Model	Accuracy	Recall	F1 Score	ROC-AUC
Logistic Regression	83.34%	14.36%	23.40%	76.90%
Random Forest	87.75%	62.83%	64.52%	93.07%
Gradient Boosting	**88.56%**	**63.97%**	**66.46%**	**93.75%**
XGBoost	87.93%	63.09%	64.95%	93.33%

"Gradient Boosting emerged as the best ML model in our study"

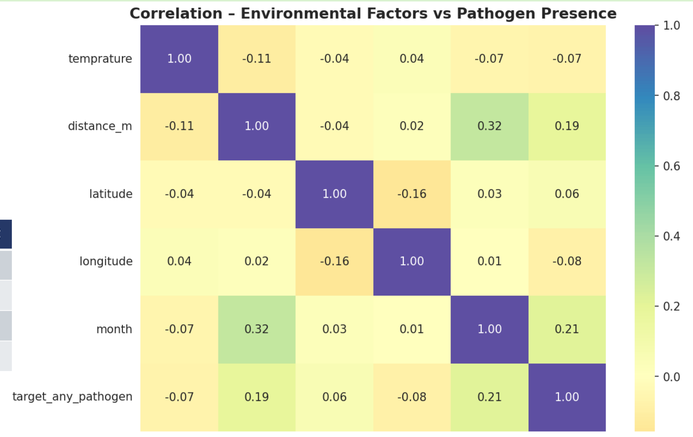


Fig 1: Correlation heatmap of environmental factors

Results

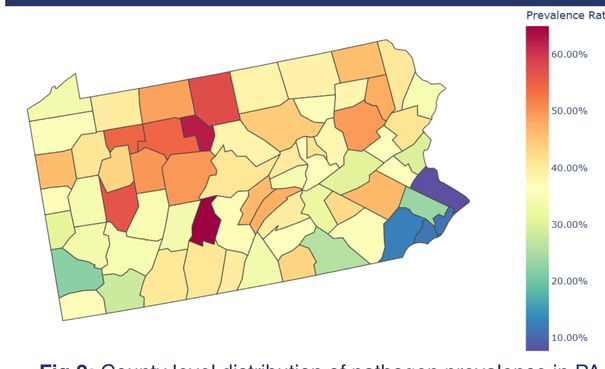


Fig 2: County level distribution of pathogen prevalence in PA

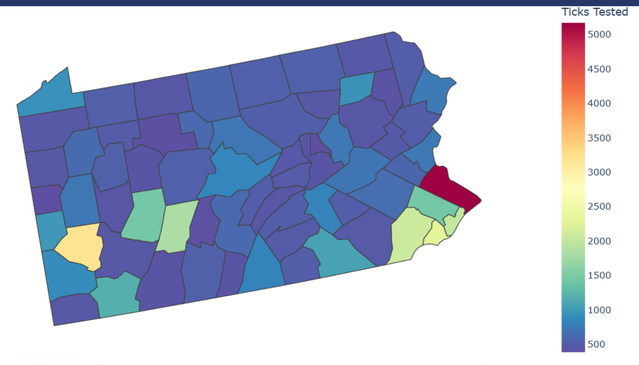


Fig 3: County level distribution of ticks tested in PA

Correlation and ML results showed that **pathogen presence** was influenced more strongly by a combination of **biological, temporal, and spatial factors** than by temperature alone. The correlation analysis indicated only weak individual relationships between environmental variables and pathogen presence, with **month** and **distance** showing the strongest positive associations, which suggests that infection risk is shaped by multiple interacting factors rather than a single dominant variable. In the life stage analysis, **adult ticks** showed higher pathogen association, while the ML models identified **nymph count**, along with geographic and seasonal variables, as **important predictors**. As shown in **Fig 2**, pathogen prevalence was not evenly distributed across PA and several counties showed clear hotspot patterns. The testing volume also varied across counties (Fig 3), but the tested areas were not always with the highest prevalence, indicating that hotspot patterns were not explained by sampling effort alone. **Gradient Boosting** performed best overall model and subsequently used for its consistent balance across **Accuracy, Recall, F1 Score, ROC AUC, and MCC**.

Conclusion

- ❑ Tick-borne disease risk in Pennsylvania is primarily influenced by biological and temporal factors rather than climate alone.
- ❑ Tick species and life stage were the strongest predictors of pathogen presence
- ❑ Adult ticks showed a higher likelihood of infection. Seasonal patterns indicate that pathogen risk varies throughout the year
- ❑ Geographic analysis revealed localized hotspots of increased risk across different regions.
- ❑ Machine learning effectively identified key patterns and improved prediction of tick-borne disease risk.

Future work

Future work will focus on incorporating additional environmental variables such as vegetation, land cover, and host species data. Expanding the dataset across broader geographic regions and longer time periods will help strengthen predictions of tick-borne disease risk. Further research can also compare different machine learning models to identify the most effective approaches. Additionally, developing higher-resolution geospatial risk maps and integrating real-time surveillance data could enhance monitoring and support early detection of high-risk areas.

References

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Acknowledgements

We acknowledge the PA DEP (Division of Vector Management) for the tick surveillance data. This work was funded in part by a Research Experience with Undergraduates (REU) grant from the Office of Research and Outreach, Penn State Harrisburg.